A MORPHISM OF SITES INDUCES POINTS AND PULLBACK PRESERVES STALKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Stalks of a pullback along a point. Same chapter, Lemma lemma-point-pushforward-sheaf. Removes the sheafification in the computation of the stalk of the pullback.
- The stalk of a presheaf as the value at the corresponding functor. Same chapter, Lemma lemma-point-functor. Identifies the composite with the stalk at the composite functor.
- The definition of a point of a site. Same chapter, Definition definition-point. Supplies the three conditions verified for the composite functor.

Source reference. Original definition 00Y5: sources/stacks-sites/pinned-original-source.tex, source lines 8783–8783. Navigation pointer invoked at proof line 8783; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00Y8: sources/stacks-sites/pinned-original-source.tex, source lines 8803–8803. Navigation pointer invoked at proof line 8803; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 0F4F: sources/stacks-sites/pinned-original-source.tex, source lines 8804–8804. Navigation pointer invoked at proof line 8804; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : \mathcal{D} \to \mathcal{C}$ be a morphism of sites given by a continuous functor $u : \mathcal{C} \to \mathcal{D}$. Let q be a point of $\mathcal{D}$ given by the functor $v : \mathcal{D} \to Sets$, see Definition 00Y5. Then the functor $v \circ u : \mathcal{C} \to Sets$ defines a point p of $\mathcal{C}$ and moreover there is a canonical identification*

$$(f^{-1}\mathcal{F})_q = \mathcal{F}_p$$

for any sheaf $\mathcal{F}$ on $\mathcal{C}$.

First proof Lemma 0.1. Note that since u is continuous and since v defines a point, it is immediate that $v \circ u$ satisfies conditions (1) and (2) of Definition 00Y5. Let us prove the displayed equality. Let $\mathcal{F}$ be a sheaf on $\mathcal{C}$. Then

$$(f^{-1}\mathcal{F})_q = (u_s\mathcal{F})_q = (u_p\mathcal{F})_q = \mathcal{F}_p$$

The first equality since $f^{-1} = u_s$, the second equality by Lemma 00Y8, and the third by Lemma 0F4F. Hence now we see that p also satisfies condition (3) of Definition 00Y5 because it is a composition of exact functors. This finishes the proof. □